

Institute of Actuaries of India

Subject SP6 – Financial Derivatives Principles

July 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

- i) Price of a 5-year 10% annual coupon bond at current rates:

$$P = 10 \sum_{i=1}^5 d_i + 100d_5$$

$$= 10(0.95238 + 0.90488 + 0.8577 + 0.81492 + 0.77796) + 100(0.77796)$$

$$= 120.875$$

Where d_i are the values of the zero-coupon bonds of length i (per nominal of 1 unit).

The value of a 5-year annual fixed-floating swap with 10% fixed coupon is the difference between a 10% 5-year bond and an FRN. But the latter is valued at 100.

Hence swap value = $120.875 - 100 = 20.875$.

[This is a receiver swap - the payer swap is of opposite sign.]

[Max 3]

- ii) Modified Duration, $M = -\frac{1}{P} \frac{\partial P}{\partial r}$

where r is the level of rates (or yields).

$$\text{Absolute Yield Sensitivity} = \frac{P_r - P_{r+\Delta r}}{\Delta r}$$

So, Absolute Yield Sensitivity of the Bond = (Modified Duration * Price of the Bond), if Δr is small and the entire yield curve shifts in parallel.

First, we need to calculate the absolute yield sensitivity of the bond and swap. Using the same technique as in (i), we get:

Bond price (shifted) = $10 (4.296) + 100 (0.77428) = 120.388$.

Hence Absolute Yield Sensitivity of the bond = $[(120.875 - 120.388) / 0.10] = 4.87$.

Similarly, the swap values (shifted) = $120.388 - 100 = 20.388$.

Hence Absolute Yield Sensitivity of the swap = $[(20.875 - 20.388) / 0.10] = 4.87$.

[Max 3]

- iii) The cash flows of the Inverse Floater are:

Coupons: Years 1 to 5: $10 - f_i$ $i = 1$ to 5 (A)

Redemption: + Year 5: 100 (B)

But the coupons on an annual fixed – floating swap are the same as in (A) and the 5-year zero coupon bond pays 100 at the end of year 5 as in (B).

Hence, the decomposition as required.

Price and absolute yield sensitivity are additive.

So, the price of the Inverse Floater = swap + 5 year zero = $20.875 + 77.796 = 98.671$.

Also, the absolute yield sensitivities = swap + 5 year zero = $4.873 + \{(77.796 - 77.428) / 0.10\}$
 $= 4.873 + 3.680 = 8.553$

So, Modified Duration of Inverse Floater = $[(8.553 * 100) / 98.671] = 8.668$, which is roughly twice the Modified Duration of the par yield bond (4.299).

[Max 7]

- iv) There are two possible alternative answers, both equally valid.

From a bond perspective:

In a bond world, the portfolio is risk managed by ensuring that modified durations match. The total duration is the sum of the individual bonds, weighted by nominal amount. Bonds with longer life will have greater convexity. Swaps are treated as a long (or short) fixed bond position and an equal short (or long) floating bond position, and their durations obtained separately. The Inverse Floater is decomposed into a swap and zero-coupon bond using the analysis in (iii) above.

OR

From a Swap perspective:

In a swap world, the risk management is performed by sensitivity analysis to the underlying yield curve. The bonds are mapped to the relevant yield curve points by taking their price sensitivity to each of the components of the curve in turn, i.e., 1 year zero, 2 year zero, etc. Swaps are decomposed into fixed and floating legs and the inverse floater is decomposed as in (iii).

[Max 2]

[15 Marks]

Solution 2:

- i) The process followed by B, the bond price, is from Ito's lemma:

$$dB = \left[\frac{\partial B}{\partial y} \mu(\alpha - y) + \frac{\partial B}{\partial t} + \frac{1}{2} \frac{\partial^2 B}{\partial y^2} \sigma^2 y^2 \right] dt + \frac{\partial B}{\partial y} \sigma y dZ \quad (2)$$

Also,

$$B = \int_0^\infty 5e^{-yt} dt = \frac{5}{y} \quad (1)$$

Thus,

$$\frac{\partial B}{\partial t} = 0; \quad \frac{\partial B}{\partial y} = -\frac{5}{y^2}; \quad \frac{\partial^2 B}{\partial y^2} = \frac{10}{y^3} \quad (2)$$

This implies

$$dB = \left[-\mu(\alpha - y) \frac{5}{y^2} + \frac{5\sigma^2}{y} \right] dt - \frac{5\sigma}{y} dZ \quad (1)$$

[Max 6]

- ii) The expected instantaneous rate at which capital gains are earned from the bond is:

$$\left[-\mu(\alpha - y) \frac{5}{y^2} + \frac{5\sigma^2}{y} \right] \quad (1)$$

The expected interest per unit time is Rs. 5. The total expected instantaneous return is:

$$\left[5 - \mu(\alpha - y) \frac{5}{y^2} + \frac{5\sigma^2}{y} \right] \quad (1)$$

When expressed as a percentage of the bond price this is:

$$\frac{\left[5 - \mu(\alpha - y) \frac{5}{y^2} + \frac{5\sigma^2}{y} \right]}{\frac{5}{y}} = y - \frac{\mu}{y}(\alpha - y) + \sigma^2 \quad (2)$$

[Max 4]

[10 Marks]

Solution 3:i) Times t or T .

[Max 1]

ii) If the option is exercised at time t , the holder of the call option receives $S(t) - X$.If it is not exercised at time t , the stock price drops to $S(t) - D$.Based on the additional information provided as a part of the question, the value of the option is $\geq S(t) - D - Xe^{-r(T-t)}$.The option should not be exercised if the value not exercised $\geq$ value exercised.

This implies that an early option exercise cannot be optimal if

$$S(t) - D - Xe^{-r(T-t)} \geq S(t) - X, \text{ i.e., } D \leq X(1 - e^{-r(T-t)}).$$

[Max 3]

iii) It is not optimal to exercise immediately prior to time t_i if $D_i \leq X[1 - \exp(-r(t_{i+1} - t_i))]$. [Max 1]

iv) One has to start by valuing the equivalent European call option.

The present value of the dividends is $\{\exp(-0.08 * 2/12) + \exp(-0.08 * 5/12)\} = 1.9540$ One can then use the Black Scholes model to value the European option using the given parameters and the time to maturity being 6 months ($T=0.5$). Here,

- $S_0 = 60 - 1.9540 = 58.0460$
- $d_1 = \{[\ln(58.046/60) + (0.08 + 0.25^2/2) * 0.5]/(0.25 * \sqrt{0.5})\} = 0.1274$
- $d_2 = d_1 - 0.25 * \sqrt{0.5} = -0.0494$
- So, $N(d_1) = 0.5507$ and $N(d_2) = 0.4803$

$$\text{Thus, } c = [(58.046 * 0.5507) - (60 * \exp(-0.08/2) * 0.4803)] = 4.277$$

$$\text{Now } 1 < 60 [1 - \exp(-0.08 * (5 - 2)/12)] = 1.188$$

So, the option should not be exercised immediately before the first ex – dividend date.

$$\text{Now, } 1 > 60 [1 - \exp(-0.08 * (6 - 5)/12)] = 0.399$$

So, it may be optimal to exercise the option immediately before the second ex – dividend date.

Therefore, one needs to revalue the option as above, but with expiry at the second ex-dividend date, i.e. $T = 5/12$ and $S_0 = 60 - \exp(-0.08 * 2/12) = 59.0134$

- $d_1 = \{[\ln(59.0134/60) + (0.08 + 0.25^2/2) * 0.41667]/(0.25 * \sqrt{0.41667})\} = 0.1845$
- $d_2 = [d_1 - (0.25 * \sqrt{0.41667})] = 0.0231$
- So, $N(d_1) = 0.5732$ and $N(d_2) = 0.5092$

$$\text{Thus, in this case, } c = \{(59.0134 * 0.5732) - (60 * 0.5092 \exp(-0.08 * 5/12))\} = 4.274$$

Black's approximation takes the maximum of these two values as the value of the American call option.

So, the determined value is 4.277

[Max 10]

[15 Marks]

Solution 4:

i) Real World projections of interest rates will allow better understanding of how interest rates could likely evolve in different scenarios. For example, there might be a need to understand whether any initial margin will cover extreme interest rate stresses.

Risk neutral projections would be helpful for understanding the complete term structure of interest rates a given point during the real – world projection. For example, there might be a need to calculate arbitrage-free prices of the securities after 10 years of the real-world projection. [Max 2]

ii) $dr(t) = a(b - r(t))dt + \sigma dz(t)$

Where...

- $r(t)$ is the short interest rate under the real – world measure
- a , b and σ are constants
- $Z(t)$ is Brownian motion

[Max 1]

iii)

- The parameters need to be calibrated based on historical movements in the short rate.
- Data can be collected using a daily, monthly, or annual frequency.
- Parameters can then be determined by either a maximum-likelihood approach or using linear regression.
- The values for μ , a , and σ will have to be subjectively determined.

[Max 2]

[5 Marks]

Solution 5:

- i) The estimation of guarantee cost requires calculating actual guarantee pay-out in various risk neutral simulation of risk drivers like equity returns, interest rate, credit default etc. These values are averaged and discounted at $t=0$ for guarantee calculation .

Risk neutral projection of interest rates, equity returns, corporate bond would be required given need of arbitrage free valuation. Various models can be used along with correction matrix to generate the scenarios and calculate the balance sheet value. Key input required are

- Interest rate model : current interest rate, swaption prices surface
- Equity & property: option pricing surface and volatility estimates
- We also need historical return index to estimate the correlation between risk

[Max 5]

- ii) We first estimate the value of fund A in different scenarios

Fund A assets			equity	
Initial asset mix		Actual exposure	50%	-50%
Equity	40%	320	480	160
Others	60%	480	480	480
	100%	800	960	640
	% Change		20%	-20%

As fund A value is passed as liability, we estimate rest of the balance sheet below.

Balance sheet t=0			
		50% rise	50% fall
Fund A	800	960	640
Fund B	400	400	400

Total asset	1200	1360	1040
Policyholder payout	1000	1200	800
Current Guarantee	100	70	130
Total Liability	1100	1270	930
Available Capital	100	90	110
	Sensitivity	-15%	

[Max 5]

- iii) The movement in free capital is counter intuitive as it reduces in equity increase scenario. This is because the liability movement due to investment return is much higher than the reduction in the guaranteed cost.

The following portfolio can be hedged by

Changing the Fund B composition to include equities such that over all Fund A+ Fund B return can match policy holder pay-out.

Use call option to hedge the guarantees by using delta hedge by using sensitivity from monte-carlo simulation.

[Max 5]

[15 Marks]

Solution 6:

- i) $V_{\text{Call}} = V_{\text{straight}} - V_{\text{callable}}$

$$V_{\text{put}} = V_{\text{puttable}} - V_{\text{Straight}}$$

[Max 2]

- ii) OAS

What Is Option-Adjusted Spread (OAS)?

The option-adjusted spread (OAS) is the measurement of the spread of a fixed-income security rate and the risk-free rate of return, which is then adjusted to take into account an embedded option. Typically, an analyst uses Treasury yields for the risk-free rate. The spread is added to the fixed-income security price to make the risk-free bond price the same as the bond.

OAS vs. Z-Spread

The Z-spread is the constant spread that makes the bond's price equal to the present value of its cash flow along each point along the Treasury curve. However, it does not include the value of the embedded options, which can have a big impact on the present value. The Z-spread is also known as the static spread because of the consistent feature.

[Max 3]

- iii) The volatility of interest rate would impact the value of the option which in turn would impact the value of the bond and OAS. As an example higher volatility would increase the value of the call option which in turn would reduce the value of the callable bond.

The option-adjusted spread is the single spread added uniformly to the one-period forward rates on the tree to produce a value or price for a bond. OAS is sensitive to interest rate volatility: The higher the volatility, the lower the OAS for a callable bond.

Assumed Level of Volatility	Value				OAS_{CALL}	OAS_{PUT}
	Calls	Puts	Callable	Putable		
High	High	High	Low	High	Low	High
Low	Low	Low	High	Low	High	Low

[Max 5]

[10 Marks]**Solution 7:**

- i) Assuming no arbitrage principle to replicate long forward we would have to borrow money and buy stock. The amount to be borrowed would be forward price of stock which assuming it is F_0 thus forward price of stock would be

$$F_0 = S_0 e^{(rT)} \text{ where } r \text{ is the risk-free rate and } t \text{ is the time period.} \quad [\text{Max 2}]$$

- ii) Forward on stock paying dividend D at time $td1$ and $td1$ and no cost of carry term T ($td1 < T$ and $Td2 > T$)
. Use 30/360 convention for valuation

Any dividend paid after the forward date would not have any impact on the price of forward.

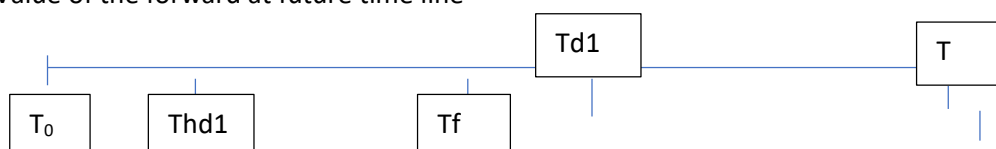
As the owner of the stock would receive the dividend that would be additional cashflow over the borrowing of the money to buy the stock. Hence at time $t=0$ assuming present value of dividend as PVD we have the following forward price

$$F = (S_0 - D \cdot e^{(-r \cdot Td1)}) \cdot e^{(rT)} \quad [\text{Max 2}]$$

- iii) Similar to above any cost paid after the expiry of the contract would not have impact on the price of the forward. Assuming present value of first payout is $HPV = H \cdot e^{(-r \cdot Thd1)}$ The forward price would be equal to

$$F = (S_0 - D \cdot e^{(-r \cdot Td1)} + H \cdot e^{(-r \cdot Thd1)}) \cdot e^{(rT)} \quad [\text{Max 3}]$$

- iv) Value of the forward at future time line



Assuming stock price at this time is S_{Tf}

We estimate the forward price at Tf and then discount the difference of the forward price to get the forward value.

$$F_0 = F = (S_0 - D \cdot e^{(-r \cdot Td1)} + H \cdot e^{(-r \cdot Thd1)}) \cdot e^{(rT)}$$

$$F_{Tf} = (S_{Tf} - D \cdot e^{(-r \cdot (Td1 - Tf))}) \cdot e^{(r \cdot (T - Tf))}$$

$$\text{Value of forward} = (F_{Tf} - F_0) e^{(-r \cdot (T - Tf))} \quad [\text{Max 4}]$$

- v) Using the formulas from part iv we get the below calculation.

S_0	100
PV Dividend	4.937889
PV holding	1.991684
F_0	98.68491
F_{Tf}	
S_t	105

PV D	4.98613
F _{tf}	100.7108
Value	2.011894

$$\text{Value of forward} = (F_{tf} - F_0)e^{(-r \cdot (T - T_f))}$$

[Max 4]

[15 Marks]**Solution 8:**

As the company payments are not in line with actual interest paid on the capital we first determine the payment amount.

Loan amount cr	100
Interest per annum(annual)	8%
total payback	103.92
payment monthly	17.32

We now estimate total money with the bank at the end of 6 month assuming the accumulation at 5%

Months	Cashflow	Months accumulation	of monthly accumulation	total accumulation	Amount
1	17.32	5	1.00407412	1.02053728	17.68
2	17.32	4	1.00407412	1.01639636	17.60
3	17.32	3	1.00407412	1.01227223	17.53
4	17.32	2	1.00407412	1.00816485	17.46
5	17.32	1	1.00407412	1.00407412	17.39
6	17.32	0	1.00407412	1	17.32
					104.99

Annual returns now can be estimated as $(104.99/100)^{1/2} - 1 = 10\%$

[5 Marks]**Solution 9:**

- i) Both warrant and call option are derivatives of the stock. Both are investment securities that an investor can use to generate a profit or leverage in an investment portfolio. Both have similar payoff on expiry.

But warrant is issued by a company while a call option is a contract between investors. Company uses warrant to raise capital.

Exchange traded call options are usually standardized, and the standards are decided by the exchanges. Warrants are non-standard and terms and conditions are decided by the company issuing the warrant.

Call option writers need to meet margin requirements. No such requirement for the company which issues the warrant.

Call options are generally short term while warrants can be issued for long term.

[Max 3]

- ii) Like a call option, warrants give the investors a non-linear payoff. When an institutional placement is done, and warrants are also given along with equity and debentures as it can enhance the returns of the investors if the company's stock does well in the future.

Since an upside potential is there, issuing warrants can help the company raise capital at more competitive rates.

Companies usually raise capital in good market conditions. Since warrants can be called for exercise after 3 years it gives additional growth capital to the company at a later date without affecting the results in the interim. Thus the company hedges itself from poor market conditions at a later date.

Since warrants are out of the money the company is in effect issuing shares at a premium to current market price. Hence it is raising capital that can be called at a price which results in lesser dilution.

Here the company is raising $180 + 300/180 \times 2150 = 3763$ Cr of capital through warrant issue. [Max 4]

- iii) Investors get exposure long term derivative that may not be available in exchange traded instruments. Even if they are available, they may not be very liquid. Warrants tend to be more liquid than long dated options (if they are available).

Some funds may have restrictions in owning exchange issued derivatives but not warrants.

Through warrants the investors can enhance their returns if their view on the stock turns out to be right.

Disadvantages:

While the upfront payout maybe low at the time of subscription the money to be invested can be quite substantial at the time of exercise. Fund size at that time maybe very different.

Substantial amount is paid as time value. If the stock doesn't perform well then the entire capital can be lost.

The investor could have taken exposure to the stock itself at a lower rate.

A holder of a warrant does not have any voting rights in a company. The investor can therefore have 'no say' in the functioning of the company, until it is converted to shares.

A warrant tracks the price movement of an underlying equity share but the investors do not get any dividend that the company may have announced for shareholders. In fact, the dividends can adversely affect the value of the warrant, unless there is a provision to adjust the strike downwards when dividend is issued.

[Max 3]

[10 Marks]
